BUILDING A USB RC METER

Episode 4 Worksheet

Measuring Capacitance with a Stopwatch

Work through these after watching the episode. Show your working; answers are on the last page. Values come from the 2014 thesis data unless stated.

Learning objectives checked

- Derive the time for an RC circuit to reach ⅔ of the supply: t = RC·ln 3
- Explain how a 555 one-shot and a comparator + timer measure capacitance
- Design the timer resolution (why 30 kΩ gives 1 overflow ≈ 1 µF)
- Explain the small-capacitor error and the non-ideal behaviour of real capacitors

Questions

1. Compute τ for R = 30 kΩ and C = 47 µF, and the time to reach ⅔ V_S.

2. Show that the time to reach ½ V_S is RC·ln 2.

3. Why does the threshold ⅔ V_S make the time independent of V_S?

4. Timer1 runs at 0.5 µs per tick. How long is one overflow, and what R makes one overflow = 2 µF?

5. Decode F100R40559 into microfarads.

6. If the firmware's reaction delay is 8 ticks, what apparent capacitance is added?

7. A 100 µF capacitor was left at 2.2 V from a previous measurement (⅔ threshold = 3.33 V). Roughly how much shorter will the next measurement be?

8. Give two reasons an electrolytic capacitor might read higher on this meter than on a DMM.

Answers

1. $\tau = 1.41$ s; $t = 1.41 \times 1.0986 \approx 1.549$ s (matches frame F47R19192).
2. $½ = 1 - e^{-t/RC} \rightarrow e^{-t/RC} = ½ \rightarrow t = RC \cdot \ln 2 \approx 0.693 \cdot RC$.
3. Both the charging voltage and the threshold scale with V_S, so V_S divides out of $1 - e^{-t/RC} = k$.
4. $65\,536 \times 0.5$ µs = 32.768 ms; $R = 32.768$ ms / (1.0986 × 2 µF) ≈ 14.9 kΩ (15 kΩ).
5. $t = 100 \times 32.768$ ms + 40 559 × 0.5 µs = 3.2971 s; $C = 3.2971 / (30\ k\Omega \times 1.0986) \approx 100.04$ µF.
6. 8 × 15.17 pF ≈ 121 pF.
7. Starting at $0.44 \cdot V_S$ instead of 0: $t = RC \cdot \ln((1 - 0.44)/(1 - 0.667)) = RC \cdot \ln(1.68) \approx 0.52 \cdot RC$ instead of $1.10 \cdot RC$, so about half the time. It reads ≈53 % low.
8. Leakage current slows the charging (reads high); the DMM may use a different test method, frequency or voltage; plus the tolerance of the 30 kΩ resistor and the ⅔ divider.

Going further

TI LMC555 CMOS timer datasheet (supply current, monostable timing):
https://www.ti.com/lit/ds/symlink/lmc555.pdf

Microchip PIC16F882/3/4/6/7 datasheet: Timer1 gate, comparator C2:
http://ww1.microchip.com/downloads/en/devicedoc/41291f.pdf

MikroElektronika: PIC Timer1 (gate control explained): https://www.mikroe.com/ebooks/pic-microcontrollers-programming-in-basic/timer1

Electrolytic leakage spec example (Nippon Chemi-Con KZH):
https://www.mouser.com/datasheet/2/420/KZHLL_e-2509011.pdf

StarFish Medical: DC bias and MLCC capacitance: https://starfishmedical.com/resource/dc-bias-on-capacitance/